

# Mathematics-I (DMA024A)



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## Mathematics II (DMA025A)

### *Course PPT*

#### Unit-II

Exact, linear and Bernoulli's equations, Equations not of first degree:  
equations solvable for p, equations solvable for y, equations solvable for x, and  
Clairaut's type

**Department of Mathematics**

**JECRC University Jaipur**

**Submitted By: Dr. Dinesh Kumar Yadav**



**JECRC Foundation**



|                |                       |                          |
|----------------|-----------------------|--------------------------|
| <b>DMA025A</b> | <b>Mathematics-II</b> | <b>3+1+0 = 4 Credits</b> |
|----------------|-----------------------|--------------------------|

### Course Contents:

#### **UNIT I: Matrices:**

Linear Systems of Equations, Linear Independence, Rank of a Matrix; Determinant, Inverse of a matrix, rank-nullity theorem, System of linear equations, Symmetric, skew-symmetric and orthogonal matrices, Determinants, Eigen values and eigenvectors, Orthogonal transformation, Diagonalization of matrices, Cayley-Hamilton Theorem.

#### **UNIT II: First order ordinary differential equations:**

Exact, linear and Bernoulli's equations. Equations not of first degree: equations solvable for p, equations solvable for y, equations solvable for x and Clairaut's type.

#### **UNIT III: Ordinary differential equations of higher orders:**

Second order linear differential equations with variable coefficients: Euler-Cauchy equations, solution by variation of parameters; Power series solutions: Legendre's equations and Legendre polynomials, Frobenius method, Bessel's equation and Bessel's functions of the first kind and their properties.

#### **UNIT IV: Complex Variable–Differentiation:**

Differentiation, Cauchy-Riemann equations, analytic functions, harmonic functions, finding harmonic conjugate, elementary analytic functions(exponential, trigonometric, logarithm) and their properties, Conformal mappings, Mobius transformations and their properties.

#### **UNIT V: Complex Variable–Integration:**

Contour integrals, Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof), Liouville's theorem and Maximum-Modulus theorem (without proof); Taylor's series, zeros of analytic functions, singularities, Laurent's series; Residues, Cauchy Residue theorem (without proof), Evaluation of definite integral involving sine and cosine, Evaluation of certain improper integrals using the Bromwich contour.

# Text/Reference Books: 1



- AICTE's Prescribed Textbook: Mathematics-II (Calculus, Ordinary Differential Equations and Complex Variable), Khanna Book Publishing Co.
- Reena Garg, Engineering Mathematics, Khanna Book Publishing Company, 2022.
- Reena Garg, Advanced Engineering Mathematics, Khanna Book Publishing Company, 2021.
- Erwin Kreyszig, Advanced Engineering Mathematics, 10<sup>th</sup> Edition, John Wiley & Sons, 2006.
- Veerarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2008.
- B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36<sup>th</sup> Edition, 2010.



## Text/Reference Books: 2



- W. E. Boyce and R. C. Di Prima, Elementary Differential Equations and Boundary Value Problems, 9<sup>th</sup> Edition, Wiley India, 2009.
- D. Poole, Linear Algebra: A Modern Introduction, 2<sup>nd</sup> Edition, Brooks/Cole, 2005.
- S. L. Ross, Differential Equations, 3<sup>rd</sup> Edition, Wiley India, 1984.
- E. A. Coddington, An Introduction to Ordinary Differential Equations, Prentice Hall India, 1995.
- E. L. Ince, Ordinary Differential Equations, Dover Publications, 1958.
- J. W. Brown and R. V. Churchill, Complex Variables and Applications, 7th Ed., McGraw Hill, 2004.
- N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.



## *Course Outcome (CO):*



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Upon successful completion of this course, the student will be able to:

- CO1: The students will learn the essential tool of matrices and linear algebra in a comprehensive manner.
- CO2: Understand the linear ordinary differential equations along with their applications.
- CO3: Will able to solve problem involving ordinary differential equations of higher orders.
- CO4: The students will learn the Complex Variable – Differentiation.
- CO5: The students will learn the Complex Variable–Integration.



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# Mapping of COs with Pos



## MAPPING COURSE OUTCOMES LEADING TO THE ACHIEVEMENT OF PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

| Course Outcome | Program Outcome |     |     |     |     |     |     |     |     |      |      |      | Program Specific Outcome |      |      |
|----------------|-----------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|--------------------------|------|------|
|                | PO1             | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1                     | PSO2 | PSO3 |
| CO1            | M               | L   |     |     | M   |     | M   |     |     |      |      | M    |                          | M    |      |
| CO2            | M               | M   |     | L   | M   |     | M   |     |     | M    | M    | M    | M                        |      |      |
| CO3            |                 | L   |     |     | M   |     |     |     |     | L    |      | M    | M                        | M    |      |
| CO4            | M               | M   |     | M   | L   |     | M   |     |     |      |      | M    |                          | M    |      |
| CO5            | M               |     |     | M   | M   |     | H   |     |     |      |      | M    | M                        | M    |      |

H=Highly Related ;M=Medium L=Low



# Contents



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- Exact,
- linear and Bernoulli's equations,
- Equations not of first degree: equations solvable for  $p$ , equations solvable for  $y$ , equations solvable for  $x$ ,
- and Clairaut's type.



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# Differential Equations of order 1 and degree 1: Exact



**Definition:** An exact differential equation of order 1 and degree 1 is a first-order equation that can be written in the form

$$M(x, y) dx + N(x, y) dy = 0, \quad \dots\dots\dots(i)$$

where  $M$  and  $N$  are functions of  $x$  and  $y$ .

The equation (i) is called **exact** if there exists a function  $F(x, y)$  such that

$$\frac{\partial F}{\partial x} = M(x, y), \quad \frac{\partial F}{\partial y} = N(x, y).$$

This happens **if and only if**

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$





# Method of Solution:



1. Compare the given equation with  $Mdx + Ndy = 0$  and find out M and N.
2. Check exactness using  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ .
3. Integrate

$$\int M dx + \int (\text{terms in } N \text{ not containing } x) dy = c'$$



# Examples.



1. Solve  $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2) = 0$ .

**Sol.** Comparing the given equation with  $M dx + N dy = 0$ , we have

$$M = x^2 - 4xy - 2y^2$$

and

$$N = y^2 - 4xy - 2x^2.$$

$$\therefore \quad \partial M / \partial y = -4x - 4y \quad \text{and} \quad \partial N / \partial x = -4y - 4x \quad \text{so that} \quad \partial M / \partial y = \partial N / \partial x.$$

Hence, the given equation is exact and hence its solution is

$$\int M dx + \int (\text{terms in } N \text{ not containing } x) dy = c'$$

[Treating  $y$  as constant]

or 
$$\int (x^2 - 4xy - 2y^2) dx + \int y^2 dy = c'$$

[Treating  $y$  as constant]

or 
$$x^3/3 - 4y \times (x^2/2) - 2y^2x + y^3/3 = c/3, \text{ taking } c' = c/3$$

or 
$$x^3 + y^3 - 6xy(x + y) = c, c \text{ being an arbitrary constant.}$$



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# Examples (Conti...)



2. Test whether the equation

$$(x + y)^2 dx - (y^2 - 2xy - x^2) dy = 0$$

is exact and hence solve it.

**Sol.** The given equation can be re-written as  $(x^2 + 2xy + y^2) dx + (x^2 + 2xy - y^2) dy = 0$ ..(1)

Comparing (1) with  $M dx + N dy = 0$ , here  $M = x^2 + 2xy + y^2$ ,  $N = x^2 + 2xy - y^2$ .

$\therefore \partial M / \partial y = 2x + 2y$  and  $\partial N / \partial x = 2x + 2y$  so that  $\partial M / \partial y = \partial N / \partial x$ .

Hence (1) is exact and hence its solution is

$$\int M dx + \int (\text{terms in } N \text{ not containing } x) dy = c'$$

[Treating y as constant]

or

$$\int (x^2 + 2xy + y^2) dx + \int (-y^2) dy = c'$$

[Treating y as constant]

or

$$x^3/3 + 2y \times (x^2/2) + y^2x - y^3/3 = c/3, \text{ taking } c' = c/3$$

or

$$x^3 + y^3 + 3xy(x + y) = c, c \text{ being an arbitrary constant.}$$



## Exercise: 1

*Solution the following differential equations:*

1.  $(x + 2y - 2) dx + (2x - y + 3) dy = 0.$

**Ans.**  $x^2 + 4xy - 4x - y^2 + 6y = c$

2.  $(2ax + by) y dx + (ax + 2by) x dy = 0.$

**Ans.**  $ayx^2 + by^2x = c$

3.  $(x^2 - ay) dx = (ax - y^2) dy.$  [Delhi Maths. (G) 1996]

**Ans.**  $x^3 - 3axy + y^3 = c$

4.  $dy/dx = (2x - y)/(x + 2y - 5).$

**Ans.**  $x^2 - xy + y^2 + 5y = c$

5.  $(x^2 + y^2 + a^2) y dy + (x^2 + y^2 - a^2) x dx = 0.$

**[S.V. University (A.P.) 1997]**

**Ans.**  $x^4 + y^4 + 2x^2y^2 + 2a^2 (y^2 - x^2) = c$

6.  $(e^y + 1) \cos x dx + e^y \sin x dy = 0.$  [Agra 2006]

**Ans.**  $(e^y + 1) \sin x = c$

7.  $x (x^2 + 3y^2) dx + y (y^2 + 3x^2) dy = 0.$

**Ans.**  $x^4 + 6x^2y^2 + y^4 = c$

8.  $(a^2 - 2xy - y^2) dx - (x + y)^2 dy = 0.$

**[Delhi B.A. (Prog) II 2011]**

**Ans.**  $a^2x - x^2y - xy^2 - (1/3) y^3 = c$

# Integrating factor:



**Definition.** If an equation of the form  $M dx + N dy = 0$  is not exact, it can always be made exact by multiplying by some function of  $x$  and  $y$ . Such a multiplier is called an integrating factor. We shall write I. F. for integrating factor.

## Rule I. By inspection.

The following list of exact differentiable should be noted carefully.

$$(i) \quad d\left(\frac{y}{x}\right) = \frac{x dy - y dx}{x^2}$$

$$(ii) \quad d\left(\frac{x}{y}\right) = \frac{y dx - x dy}{y^2}$$

$$(iii) \quad d\left(\frac{y^2}{x}\right) = \frac{2xy dy - y^2 dx}{x^2}$$

$$(iv) \quad d\left(\frac{x^2}{y}\right) = \frac{2yx dx - x^2 dy}{y^2}$$

$$(v) \quad d\left(\frac{y^2}{x^2}\right) = \frac{2x^2 y dy - 2xy^2 dx}{x^4}$$

$$(vi) \quad d\left(\frac{x^2}{y^2}\right) = \frac{2y^2 x dx - 2yx^2 dy}{y^4}$$

$$(vii) \quad d[\log(xy)] = \frac{x dy + y dx}{xy}$$

$$(viii) \quad d(xy) = x dy + y dx$$





## Integrating factor (Continu...):

$$(ix) \quad d \left( \tan^{-1} \frac{y}{x} \right) = \frac{x dy - y dx}{x^2 + y^2}$$

$$(xi) \quad d \left[ \log \left( \frac{y}{x} \right) \right] = \frac{x dy - y dx}{xy}$$

$$(xiii) \quad d \left[ \frac{1}{2} \log (x^2 + y^2) \right] = \frac{x dx + y dy}{x^2 + y^2}$$

$$(xv) \quad d \left( \frac{e^x}{y} \right) = \frac{y e^x dx - e^x dy}{y^2}$$

$$(x) \quad d \left( \tan^{-1} \frac{x}{y} \right) = \frac{y dx - x dy}{x^2 + y^2}$$

$$(xii) \quad d \left[ \log \left( \frac{x}{y} \right) \right] = \frac{y dx - x dy}{xy}$$

$$(xiv) \quad d \left( -\frac{1}{xy} \right) = \frac{x dy + y dx}{x^2 y^2}$$

$$(xvi) \quad d (\sin^{-1} xy) = \frac{x dy + y dx}{(1 - x^2 y^2)^{1/2}}$$





## Examples.

**Ex. 1.** Solve  $y dx - x dy + (1 + x^2) dx + x^2 \sin y dy = 0$ .

[Allahabad 1996]

**Sol.** Dividing each term of the given equation by  $x^2$ , we get

$$\frac{y dx - x dy}{x^2} + \frac{1 + x^2}{x^2} dx + \sin y dy = 0 \quad \text{or} \quad -\frac{x dy - y dx}{x^2} + \left(\frac{1}{x^2} + 1\right) dx + \sin y dy = 0$$

or 
$$-d(y/x) + (1 + 1/x^2) dx + \sin y dy = 0.$$

Integrating, 
$$-(y/x) + x - (1/x) - \cos y = c$$

or 
$$-y + x^2 - 1 - x \cos y = cx, \text{ where } c \text{ is an arbitrary constant.}$$

**Ex. 2.** Solve  $y(2xy + e^x) dx = e^x dy$ .

[Agra 1995; Lucknow 1998]

**Sol.** Re-writing,  $2xy^2 dx + y e^x dx - e^x dy = 0$

or 
$$2x dx + \frac{y e^x dx - e^x dy}{y^2} = 0 \quad \text{or} \quad 2x dx + d\left(\frac{e^x}{y}\right) = 0.$$

Integrating, 
$$x^2 + e^x/y = c \quad \text{or} \quad yx^2 + e^x = cy.$$

## Exercise: 2



*Solve the following differential equations:*

1.  $e^y dx + (xe^y + 2y) dy = 0$ . **Ans.**  $xe^y + y^2 = c$
  2.  $x dx + y dy + (x^2 + y^2) dy = 0$ . **Ans.**  $x^2 + y^2 = ce^{-2y}$
  3.  $x dy - y dx = (x^2 + y^2) dx$ . **Ans.**  $\tan^{-1}(y/x) = x + c$
  4.  $y dx - x dy + \log x dx = 0$ . **Ans.**  $cx + y + \log x + 1 = 0$
  5.  $e^{2y} dx + 2(xe^{2y} - y) dy = 0$ . **Ans.**  $xe^{2y} - y^2 = c$
  6.  $y(2x^2y + e^x) dx - (e^x + y^3) dy = 0$ . **Ans.**  $4x^3y - 3y^3 + 6e^x = 6cy$
  7.  $(x^3 e^x - my^2) dx + mxy dy = 0$ . **Ans.**  $e^x + (my^2)/(2x^2) = c$
  8.  $x dy - y dx = xy^2 dx$ . **Ans.**  $yx^2 + 2x = 2cy$
  9.  $y(axy + e^x) dx - e^x dy = 0$ . **Ans.**  $ax^2y + 2e^x = cy$
  10.  $\{y + \cos y + 1/(2\sqrt{x})\} dx + (x - x \sin y - 1) dy = 0$ . **Ans.**  $xy + x \cos y + x^{1/2} - y = c$
  11.  $(x^2 + y^2 - a^2) x dx + (x^2 - y^2 - b^2) y dy = 0$ . **Ans.**  $(x^2 - a^2)^2 - (y^2 + b^2)^2 + 2x^2y^2 = c$
  12.  $a(x dy + 2y dx) = xy dy$ . **Ans.**  $a \log(yx^2) - y = c$
- [**Hint.** Divide by  $xy$ , i.e., take  $1/(xy)$  as an I.F.]
13.  $dx + y dy = m(x dy - y dx)$ . **[Delhi Maths(H) 2000]**



## Rule II.

***If the given equation  $Mdx + Ndy = 0$  is homogeneous and  $(Mx + Ny) \neq 0$ , then  $\frac{1}{(Mx+Ny)}$  is an integrating factor.***

**Ex. 1.** Solve  $(x^2y - 2xy^2) dx - (x^3 - 3x^2y) dy = 0$ . [Delhi Maths (G) 1994; Garhwal 2010]

**Sol.** Given  $(x^2y - 2xy^2) dx - (x^3 - 3x^2y) dy = 0$ . ... (1)

Clearly (1) is a homogeneous differential equation. Comparing (1) with  $M dx + N dy = 0$ ,

$M = x^2y - 2xy^2$  and  $N = -(x^3 - 3x^2y)$  ....(2)

$\therefore Mx + Ny = x(x^2y - 2xy^2) - y(x^3 - 3x^2y) = x^2y^2 \neq 0$ ,

showing that I.F. of (1)  $= 1/(Mx + Ny) = 1/(x^2y^2)$ . On multiplying (1) by  $1/(x^2y^2)$ ,

$(1/y - 2/x) dx - (x/y^2 - 3/y) dy = 0$ , which is exact where solution is

$$\int \{(1/y) - (2/x)\} dx + \int (3/y) dy = 0 \quad \text{or} \quad (x/y) - 2 \log x + 3 \log y = \log c$$

[Treating y as constant]

$$\text{or} \quad \log y^2 - \log x^2 - \log c = -x/y \quad \text{or} \quad \log (y^2/cx^2) = -x/y$$

$$\text{or} \quad y^2 = cx^2 e^{-x/y}, \text{ where } c \text{ is an arbitrary constant.}$$



## Rule III.

***If the given equation  $Mdx + Ndy = 0$  is of the form  $f_1(xy)ydx + f_2(xy)x dy$ , then  $\frac{1}{(Mx - Ny)}$ ,  $(Mx - Ny) \neq 0$  is an integrating factor.***

**Ex. 1.** Solve  $(xy \sin xy + \cos xy) y dx + (xy \sin xy - \cos xy) x dy = 0$ .

[Purvanchal 1996; Kanpur 1993; Lucknow 1993, 1997]

**Sol.** Given  $(xy \sin xy + \cos xy) y dx + (xy \sin xy - \cos xy) x dy = 0$  ... (1)

Comparing (1) with  $M dx + N dy = 0$ , we have

$M = y (xy \sin xy + \cos xy)$  and  $N = x (xy \sin xy - \cos xy)$ ,

showing that (1) is of the form  $f_1(xy) y dx + f_2(xy) x dy = 0$ .

Again,  $Mx - Ny = xy (xy \sin xy + \cos xy) - xy (xy \sin xy - \cos xy)$

$\therefore Mx - Ny = 2xy \cos xy \neq 0$ . Hence I.F. of (1)  $= 1/(Mx - Ny) = 1/(2xy \cos xy)$ .







## Rule III.

On multiplying (1) by  $1/(2xy \cos xy)$ , we have

$$(1/2) \times (y \tan xy + 1/x) dx + (1/2) \times (x \tan xy - 1/y) dy = 0 \quad \dots (2)$$

which must be exact and so by the usual rule, solution of (2) is

$$\int \{(1/2) \times (y \tan xy + 1/x)\} dx + \int (-1/2y) dy = (1/2) \times \log c$$

[Treating  $y$  as constant]

$$\text{or} \quad (1/2) \times (\log \sec xy + \log x) - (1/2) \times \log y = (1/2) \log c$$

$$\text{or} \quad \log \sec xy + \log (x/y) = \log c \quad \text{or} \quad (x/y) \sec xy = c.$$

**Ex. 2.** Solve  $y(1 + xy) dx + x(1 - xy) dy = 0$ .

[I.A.S. (Prel.) 2006; Meerut 1993;  
G.N.D.U. Amritsar 2010]

$$\text{Sol. Given} \quad (1 + xy) y dx + (1 - xy) x dy = 0. \quad \dots (1)$$

Comparing (1) with  $M dx + N dy = 0$ ,  $M = (1 + xy) y$  and  $N = (1 - xy) x$ , showing that (1) is of the form  $f_1(xy) y dx + f_2(xy) x dy = 0$ .

Again,  $Mx - Ny = xy(1 + xy) - xy(1 - xy) = 2x^2y^2 \neq 0$ ,  
showing that I.F. of (1)  $= 1/(Mx - Ny) = 1/(2x^2y^2)$ .

On multiplying (1) by  $1/(2x^2y^2)$ , we have

$$\frac{1}{2} \left( \frac{1}{x^2y} + \frac{1}{x} \right) dx + \frac{1}{2} \left( \frac{1}{xy^2} - \frac{1}{y} \right) dy = 0,$$



## Rule III.

which must be exact and so by the usual rule, solution of (2) is

$$\int \left( \frac{1}{2x^2y} + \frac{1}{2x} \right) dx + \int \left( -\frac{1}{2y} \right) dy = \frac{1}{2} \log c \quad \text{or} \quad \frac{1}{-2xy} + \frac{1}{2} \log x - \frac{1}{2} \log y = \frac{1}{2} \log c$$

[Treating  $y$  as constant]

$$\text{or} \quad \log(x/y) - \log c = 1/(xy) \quad \text{or} \quad \log(x/cy) = 1/(xy) \quad \text{or} \quad x = cy e^{1/(xy)}.$$



## Exercise: 3



*Solve the following differential equations:*

1.  $(x^2y^2 + xy + 1)y \, dx + (x^2y^2 - xy + 1)x \, dy = 0.$

**Ans.**  $xy - (1/xy) + \log(x/y) = c$

2.  $(x^4y^4 + x^2y^2 + xy)y \, dx + (x^4y^4 - x^2y^2 + xy)x \, dy = 0.$

**Ans.**  $(1/2) \times x^2y^2 - (1/xy) + \log(x/y) = c$

3.  $y(1 - xy) \, dx - x(1 + xy) \, dy = 0.$  [Agra 1994; I.A.S. 1969]

**Ans.**  $\log(x/y) - xy = c$

4.  $(xy^2 + 2x^2y^3) \, dx + (x^2y - x^3y^2) \, dy = 0.$  [I.A.S. 2004]

**Ans.**  $\log(x^2/y) - (1/xy) = c$



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## Rule IV.



*If  $\frac{1}{N} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$  is a function of  $x$  alone say  $f(x)$ , then  $e^{\int f(x)dx}$  is an integrating factor of  $Mdx + Ndy = 0$ .*

**Ex. 1.** Solve  $(x^2 + y^2 + x) dx + xy dy = 0$ .

[Delhi B.Sc. (Prog) II, 2009]

**Sol.** Given  $(x^2 + y^2 + x) dx + xy dy = 0$ . ... (1)

Comparing (1) with  $M dx + N dy = 0$ , here  $M = x^2 + y^2 + x$  and  $N = xy$ .

Here  $\partial M / \partial y = 2y$  and  $\partial N / \partial x = y$ . So  $\partial M / \partial y \neq \partial N / \partial x$ . We have

$$\frac{1}{N} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{1}{xy} (2y - 1) = \frac{1}{x}, \text{ which is a function of } x \text{ alone.}$$

$$\therefore \text{I.F. of (1)} = e^{\int (1/x) dx} = e^{\log x} = x.$$

Multiplying (1) by  $x$ , we have  $(x^3 + xy^2 + x^2) dx + x^2y dy = 0$ ,

which must be exact equation and so its solution as usual is

$$\int (x^3 + xy^2 + x^2) dx = (1/6) \times c \quad \text{or} \quad (1/4) \times x^4 + (1/2) \times x^2 y^2 + (1/3) \times x^3 = c/6$$

[Treating  $y$  as constant]

or  $3x^4 + 6x^2y^2 + 4x^3 = c$ , where  $c$  is an arbitrary constant.

## Rule IV.



**Ex. 2.** Solve  $(y + y^3/3 + x^2/2) dx + (1/4) \times (x + xy^2) dy = 0$ .

[Allahabad 1994]

**Sol.** Given  $(y + y^3/3 + x^2/2) dx + (1/4) \times (x + xy^2) dy = 0$ . ... (1)

Comparing (1) with  $M dx + N dy = 0$ ,  $M = y + y^3/3 + x^2/2$  and  $N = (1/4) \times (x + xy^2)$ .

Here  $\partial M / \partial y = 1 + y^2$  and  $\partial N / \partial x = (1/4) \times (1 + y^2)$ .

$$\therefore \frac{1}{N} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{4}{x(1+y^2)} \left\{ (1+y^2) - \frac{1}{4}(1+y^2) \right\} = \frac{4}{x} \left( 1 - \frac{1}{4} \right) = \frac{3}{x},$$

which is a function of  $x$  alone. So I.F. =  $e^{\int (3/x) dx} = e^{3 \log x} = e^{\log x^3} = x^3$ .

Multiplying (1) with  $x^3$ , we have

$\{x^3y + (1/3)x^3y^3 + (1/2)x^5\} dx + (1/4)(x^4 + x^4y^2) dy = 0$  whose solution as usual is

$$\int \{x^3y + (1/3)x^3y^3 + (1/2)x^5\} dx = c/12 \quad \text{or} \quad (1/4) \times x^4y + (1/12) \times x^4y^3 + (1/12) \times x^6 = c/6$$

[Treating  $y$  as constant]

or  $3x^4y + x^4y^3 + x^6 = c$ , where  $c$  is an arbitrary constant.



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## Exercise. 4



*Solve the following differential equations:*

1.  $(x^2 + y^2 + 2x) dx + 2y dy = 0.$

**Ans.**  $e^x (x^2 + y^2) = c$

2.  $(x^3 - 2y^2) dx + 2xy dy = 0.$

**Ans.**  $x + (y^2/x^2) = c$

3.  $(x^2 + y^2) dx - 2xy dy = 0.$  (Pune 2010)

**Ans.**  $x^2 - y^2 = cx$

4.  $(x^2 + y^2 + 1) dx - 2xy dy = 0.$  (Delhi B.Sc. (Prog.) II 2008)

**Ans.**  $x^2 - 1 - y^2 = cx$

5.  $(x^2 + y^2 + 1) dx + x(x - 2y) dy = 0.$

**Ans.**  $x + y - (y^2 + 1)/x = c$

6.  $(5xy + 4y^2 + 1)dx + (x^2 + 2xy)dy = 0$  [Delhi B.A. (Prog.) II 2010] **Ans.**  $x^5 y + x^4 y^2 + x^4 / 4 = c$



# Rule V.



If  $\frac{1}{M} \left( \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)$  is function of  $y$  alone, say  $f(y)$ , then  $e^{\int f(y) dy}$  is an integrating factor of  $M dx + N dy = 0$ .

**Ex. 1.** Solve  $(2xy^4e^y + 2xy^3 + y) dx + (x^2y^4e^y - x^2y^2 - 3x) dy = 0$ . ... (1)

**Sol.** Comparing (1) with  $M dx + N dy = 0$ , we get

$$M = 2xy^4e^y + 2xy^3 + y \quad \text{and} \quad N = x^2y^4e^y - x^2y^2 - 3x. \dots (2)$$

$$\text{Here } \frac{\partial M}{\partial y} = 8xy^3e^y + 2xy^4e^y + 6xy^2 + 1 \quad \text{and} \quad \frac{\partial N}{\partial x} = 2xy^4e^y - 2xy^2 - 3.$$

$$\therefore \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = -4(2xy^3e^y + 2xy^2 + 1) = -\frac{4}{y}(2xy^4e^y + 2xy^3 + y) = -\frac{4M}{y}$$

$$\Rightarrow \frac{1}{M} \left( \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) = -\frac{4}{y}, \text{ which is a function of } y \text{ alone.}$$

$$\Rightarrow \text{I.F. of (1)} = e^{\int (-4/y) dy} = e^{-4 \log y} = (1/y^4).$$

Multiplying (1) by  $1/y^4$ , we have

$$\{2xe^y + (2x/y) + (1/y^3)\} dx + \{x^2e^y - (x^2/y^2) - 3(x/y^4)\} dy = 0 \text{ whose solution as usual is}$$

$$\int \{2xe^y + (2x/y) + (1/y^3)\} dx = c \quad \text{or} \quad x^2e^y + (x^2/y) + (x/y^3) = c.$$

[Treating  $y$  as constant]

# Rule V.



**Ex. 2.** Solve  $(xy^2 - x^2) dx + (3x^2y^2 + x^2y - 2x^3 + y^2) dy = 0$ .

**Sol.** Given  $(xy^2 - x^2) dx + (3x^2y^2 + x^2y - 2x^3 + y^2) dy = 0$ . ... (1)

Comparing (1) with  $M dx + N dy = 0$ ,  $M = xy^2 - x^2$ ,  $N = 3x^2y^2 + x^2y - 2x^3 + y^2$ .

$\therefore \partial M / \partial y = 2xy$  and  $\partial N / \partial x = 6xy^2 + 2xy - 6x^2$ .

$$\therefore \frac{1}{M} \left( \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) = \frac{1}{xy^2 - x^2} \{ (6xy^2 + 2xy - 6x^2) - 2xy \} = \frac{6x(y^2 - x)}{x(y^2 - x)} = 6,$$

which being a constant can be treated as a function of  $y$  alone.

$\therefore$  I.F. of (1) =  $e^{\int 6dy} = e^{6y}$ . Multiplying (1) by  $e^{6y}$ , we have

$e^{6y} (xy^2 - x^2) dx + e^{6y} (3x^2y^2 + x^2y - 2x^3 + y^2) dy = 0$  whose solution is

$$\int e^{6y} (xy^2 - x^2) dx + \int e^{6y} y^2 dy = c$$

[Treating  $y$  as constant]

or 
$$e^{6y} \left[ (1/2)x^2y^2 - (1/3)x^3 \right] + y^2(1/6)e^{6y} - \int (2y)(1/6)e^{6y} dy = c$$

or 
$$e^{6y} \left( \frac{1}{2}x^2y^2 - \frac{1}{3}x^3 \right) + \frac{1}{6}y^2e^{6y} - \frac{1}{3} \left[ y \times \frac{1}{6}e^{6y} - \int \left( 1 \times \frac{1}{6}e^{6y} \right) dy \right] = c$$

or 
$$e^{6y} \left( \frac{1}{2}x^2y^2 - \frac{1}{3}x^3 \right) + \frac{1}{6}y^2e^{6y} + \frac{1}{3} \left[ \frac{1}{6}ye^{6y} - \frac{1}{36}e^{6y} \right] = c$$

or 
$$e^{6y} \left( \frac{1}{2}x^2y^2 - \frac{1}{3}x^3 + \frac{1}{6}y^2 - \frac{1}{18}y + \frac{1}{108} \right) = c. \text{ } c \text{ being an arbitrary constant.}$$



## Exercise. 5

*Solve the following differential equations:*

1.  $(xy^3 + y) dx + 2(x^2y^2 + x + y^4) dy = 0.$

**(Delhi 2009) Ans.**  $3x^2y^4 + 6xy^2 + 2y = c$

2.  $(2xy^2 - 2y) dx + (3x^2y - 4x) dy = 0.$

**[Delhi B.A (Prog) II 2011]**

**Ans.**  $x^2y^3 + 2xy^2 = c$

3.  $(y^4 + 2y) dx + (xy^3 + 2y^4 - 4x) dy = 0$

**[Delhi Maths (H) 2007, 08;**



## Rule VI.



*If the given equation  $M dx + N dy = 0$ , is of the form  $x^\alpha y^\beta (my dx + nx dy) = 0$ , then its integrating factor is  $x^{km-1-\alpha} y^{kn-1-\beta}$ , where  $k$  can have any value.*

**Ex.1** Solve  $(y^2 + 2x^2y) dx + (2x^3 - xy) dy = 0$ .

**Sol.** Re-writing (1) in the standard form

$$x^\alpha y^\beta (my dx + nx dy) x^{\alpha'} y^{\beta'} (m'y dx + n'x dy) = 0, \quad \dots (2)$$

we have

$$y (y dx - x dy) + x^2 (2y dx + 2x dy) = 0. \quad \dots (3)$$

Comparing (2) and (3), we have

$$\alpha = 0, \quad \beta = 1, \quad m = 1, \quad n = -1; \quad \alpha' = 2, \quad \beta' = 0, \quad m' = 2, \quad n' = 2.$$

Hence, the I.F. for the first term on L.H.S. of (3) is  $x^{k-1} y^{-k-1-1}$ , i.e.,  $x^{k-1} y^{-k-2} \dots (4)$

and the I.F. for the second term on L.H.S. of (3) is  $x^{2k'-1-2} y^{2k'-1}$ , i.e.,  $x^{2k'-3} y^{2k'-1} \dots (5)$

For the integrating factors (4) and (5) to be identical, we have

$$\begin{aligned} k-1 &= 2k'-3 & \text{and} & & -k-2 &= 2k'-1 \\ \Rightarrow k-2k' &= -2 & \text{and} & & k+2k' &= -1 \end{aligned} \Rightarrow k = -3/2 \quad \text{and} \quad k' = 1/4 \quad \dots (6)$$



## Rule VI.



Substituting the value of  $k$  in (4) or  $k'$  in (5), the integrating factor of (3) or (1) is  $x^{-5/2} y^{-1/2}$ .

Multiplying (1) by  $x^{-5/2} y^{-1/2}$ , we have

$$(x^{-5/2} y^{3/2} + 2x^{-1/2} y^{1/2}) dx + (2x^{1/2} y^{-1/2} - x^{-3/2} y^{1/2}) dy = 0,$$

which must be exact and so by the usual rule its solution is given by

$$\frac{x^{-3/2} y^{3/2}}{(-3/2)} + \frac{2x^{1/2} y^{1/2}}{(1/2)} = \frac{2C}{3} \quad \text{or} \quad 6x^{1/2} y^{1/2} - x^{-3/2} y^{3/2} = C.$$

**Ex. 2** Solve  $(y^2 + 2x^2y) dx + (2x^3 - xy) dy = 0$ . **[Allahabad 1993; Lucknow 1993]**

**Sol.** Given  $(y^2 + 2x^2y) dx + (2x^3 - xy) dy = 0$ . ... (1)

Re-writing (1),  $(y^2 dx - xy dy) + (2x^2y dx + 2x^3 dy) = 0$

or  $y(y dx - x dy) + x^2(2y dx + 2x dy) = 0$ . **[Delhi 2009]**

which is of the form  $x^\alpha y^\beta (my dx + nx dy) + x^{\alpha'} y^{\beta'} (m'y dx + n'x dy) = 0$ .

So, let  $x^h y^k$  be an I.F. (1). Multiplying (1) by  $x^h y^k$ , we have

$$(x^h y^{k+2} + 2x^{h+2} y^{k+1}) dx + (2x^{h+3} y^k - x^{h+1} y^{k+1}) dy = 0, \quad \text{... (2)}$$

which must be exact. Comparing (2) with  $M dx + N dy = 0$ , we get

$$M = x^h y^{k+2} + 2x^{h+2} y^{k+1} \quad \text{and} \quad N = 2x^{h+3} y^k - x^{h+1} y^{k+1}.$$

## Rule VI.



Since (2) is exact, we must have

$$\partial M / \partial y = \partial N / \partial x,$$

$$\text{i.e.,} \quad (k+2) x^h y^{k+1} + 2(k+1) x^{h+2} y^k = 2(h+3) x^{h+2} y^k - (h+1) x^h y^{k+1}.$$

Now equating the coefficients of  $x^h y^{k+1}$  and  $x^{h+2} y^k$ , we get

$$k+2 = -(h+1) \quad \text{and} \quad 2(k+1) = 2(h+3),$$

$$\text{i.e.,} \quad h+k = -3 \quad \text{and} \quad h-k = -2 \quad \text{giving} \quad h = -(5/2), \quad k = -(1/2)$$

$\therefore$  I.F. =  $x^{-5/2} y^{-1/2}$ . Multiplying (1) by I.F.  $x^{-5/2} y^{-1/2}$ , we get

$$(x^{-5/2} y^{3/2} + 2x^{-1/2} y^{1/2}) dx + (2x^{1/2} y^{-1/2} + x^{-3/2} y^{1/2}) dy = 0$$

which must be exact. For this new equation, as usual its solution is

$$\int (x^{-5/2} y^{3/2} + 2x^{-1/2} y^{1/2}) dx = c \quad \text{or} \quad -\frac{2}{3} x^{-3/2} y^{3/2} + 4x^{1/2} y^{1/2} = c.$$

[Treating  $y$  as constant]



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## Exercise. 6

*Solve the following differential equations:*

1.  $(2x^2y - 3y^4) dx + (3x^3 + 2xy^3) dy = 0.$

**Ans.**  $12x^{-10/13} y^{15/13} + 5x^{-36/13} y^{24/13} = c$

2.  $(3x + 2y^2) y dx + 2x (2x + 3y^2) dy = 0.$

**Ans.**  $x^3y^4 + x^2y^6 = c$

3.  $x (3y dx + 2x dy) + 8y^4 (y dx + 3x dy) = 0.$

**Ans.**  $x^2y^3 (x + 4y^4) = c$

4.  $x (4y dx + 2x dy) + y^3 (3y dx + 5x dy) = 0.$  [Delhi Maths (G) 1999]

**Ans.**  $x^4y^3 + x^3y^5 = c$

5.  $xy^3 (y dx + 2x dy) + (3y dx + 5x dy) = 0.$

**Ans.**  $x^3y^5 (xy^3 + 4) = c$

6.  $(8x^2y^3 - 2y^4) dx + (5x^3y^2 - 8xy^3) dy = 0$  by first finding integrating factor of the form  $x^h y^k$ .

[Delhi. B.Sc. (Prog) II 2011]

**Ans.**  $x^{8/3} y^{5/3} - x^{2/3} y^{8/3} = c$



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# Linear differential equation: Type-1

**Definition.** A first order differential equation is called linear if it can be written in the form

$$\frac{dy}{dx} + Py = Q, \quad \dots(1)$$

where P and Q are constants or functions of x alone (i.e., not of y).

**Working rule for solving linear equations.** First put the given equation in the standard form (1). Next find an integrating factor (I.F.) by using formula

$$\text{I.F.} = e^{\int P dx} \quad \dots (5)$$

Two formulas  $e^{m \log A} = A^m$  and  $e^{-m \log A} = 1/A^m$  will be often used in simplifying I.F.

Lastly, the required solution is obtained by using the result

$$y \times (\text{I.F.}) = \int [Q \times (\text{I.F.})] dx + c, \text{ where } c \text{ is an arbitrary constant.} \quad \dots (6)$$



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## Linear differential equation: Type-2

**Definition.** A first order differential equation is called linear if it can be written in the form

$$\frac{dx}{dy} + P_1x = Q_1, \quad \dots(2)$$

where  $P_1$  and  $Q_1$  are constants or functions of  $y$  alone (i.e., not of  $x$ ).

**Working rule for solving linear equations.** First put the given equation in the standard form (2). Next find an integrating factor (I.F.) by using formula

as follows.

$$\text{I.F.} = e^{\int P_1 dy}$$

and the required solution is

$$x \times (\text{I.F.}) = \int [Q_1 \times (\text{I.F.})] dy + c.$$



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# Examples.



**Ex.1.** Solve  $x \cos x \frac{dy}{dx} + y(x \sin x + \cos x) = 1$ .

**Solu.** Re-writing given equation, we have

$$\frac{dy}{dx} + \left( \tan x + \frac{1}{x} \right) y = \frac{\sec x}{x} \quad \dots (1)$$

$$\text{I.F. of (1)} = e^{\int (\tan x + 1/x) dx} = e^{\log \sec x + \log x} = e^{\log x \sec x} = x \sec x.$$

Hence the required solution is  $yx \sec x = \int \sec^2 x dx + c$ ,

or  $yx \sec x = \tan x + c$ ,  $c$  being arbitrary constants.



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# Examples (Continu...).



**Ex. 2. (a)** Solve  $(1 - x^2)(dy/dx) + 2xy = x\sqrt{(1 - x^2)}$ .

[Kerala 2001]

**(b)** solve  $(1 - x^2)(dy/dx) + 2xy = x\sqrt{1 - x^2}$ ,  $y(0) = 1$

[Delhi Maths (Prog) 2007]

**Sol.** The given equation is 
$$\frac{dy}{dx} + \frac{2x}{1 - x^2} y = \frac{x}{(1 - x^2)^{1/2}} \quad \dots (1)$$

Comparing (1) with  $dy/dx + Py = Q$ , here  $P = 2x/(1 - x^2)$

Here  $\int P dx = \int \frac{2x}{1 - x^2} dx = -\log(1 - x^2)$  hence I.F. of (1) =  $e^{\int P dx} = \frac{1}{1 - x^2}$

So the required solution is

$$\frac{y}{1 - x^2} = \int \frac{x}{\sqrt{(1 - x^2)}} \times \frac{1}{1 - x^2} dx = -\frac{1}{2} \int t^{-3/2} dt + c, \text{ putting } 1 - x^2 = t \text{ and } -2x dx = dt$$

$$\text{or } \frac{y}{1 - x^2} = t^{-1/2} + c = c + \frac{1}{\sqrt{t}} \quad \text{or} \quad \frac{y}{1 - x^2} = \frac{1}{(1 - x^2)^{1/2}} + c, \text{ as } t = 1 - x^2 \quad \dots (2)$$

(b) First do upto equation (2) as in Ex. 2(a). Putting  $x = 0$  and  $y = 1$  in (2), we have  $1 = 1 + c$  so that  $c = 0$ . Hence (2) becomes

$$y/(1 - x^2) = 1/(1 - x^2)^{1/2} \quad \text{or} \quad y = (1 - x^2)^{1/2}$$

## Examples (Continu...).



**Ex. 3.** Solve  $(x + 2y^3) \left( \frac{dy}{dx} \right) = y$ .

**Sol.** Here it is possible to put the equation in form  
where  $P_1$  and  $Q_1$  are function of  $y$  or constants

$$dx/dy + P_1 x = Q_1.$$

Thus, we have  $\frac{dx}{dy} = \frac{x + 2y^3}{y}$ , or  $\frac{dx}{dy} - \frac{1}{y} x = 2y^2$ . ... (1)

For (1),  $\int P_1 dy = -\int (1/y) dy = -\log y$  so I.F. of (1) =  $e^{-\log y} = 1/y$ .

Hence, the required solution is  $x/y = \int 2y^2 \cdot (1/y) dx + c$ .

or  $x/y = y^2 + c$ , where  $c$  is an arbitrary constant.



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## Exercise. 7

*Solve the following differential equations:*

1.  $(1 + x^2) (dy/dx) + y = e^{\tan^{-1} x}$ .  
**Ans.**  $ye^{\tan^{-1} x} = (1/2) \times e^{2 \tan^{-1} x} + c$
2.  $(dy/dx) + y \cot x = 2 \cos x$ . [Bangalore 1994]  
**Ans.**  $y \sin x = -(1/2) \times \cos 2x + c$
3.  $x \log x (dy/dx) + y = 2 \log x$ . [Delhi (Maths (G) 2005]  
**Ans.**  $y \log x = c + (\log x)^2$
4.  $dy/dx = y \tan x - 2 \sin x$ .  
**Ans.**  $y = \cos x + c \sec x$
5.  $(dy/dx) + 2y \tan x = \sin x$ , given that  $y = 0$  when  $x = \pi/3$ .  
**Ans.**  $y = \cos x - 2 \cos^2 x$
6.  $\cos^2 x (dy/dx) + y = \tan x$ .  
**Ans.**  $y = \tan x - 1 + ce^{\tan^{-1} x}$
7.  $dy/dx + 2 (y/x) = \sin x$ .  
**Ans.**  $yx^2 = c - x^2 \cos x + 2x \sin x + 2 \cos x$
8.  $(2x - 10y^3) (dy/dx) + y = 0$ .  
**Ans.**  $xy^2 = c + 2y^5$
9.  $(x \log x) (dy/dx) + y = 2 \log x$ . [Delhi Maths. (G) 1996]  
**Ans.**  $y \log x = c + (\log x)^2$
10.  $\cos x (dy/dx) + y = \sin x$  or  $(dy/dx) + y \sec x = \tan x$ .  
**Ans.**  $y (\sec x + \tan x) = \sec x + \tan x - x + c$
11.  $(1 + y^2) + (x - e^{\tan^{-1} y}) (dy/dx) = 0$ .  
**Ans.**  $xe^{\tan^{-1} y} = (1/2) \times e^{2 \tan^{-1} y} + c$
12.  $x (dy/dx) - y = 2x^2 \operatorname{cosec} x$ . [Kanpur 1996]  
**Ans.**  $y = cx + x \log (\tan x)$
13.  $x^2 (x^2 - 1) (dy/dx) + x (x^2 + 1) y = x^2 - 1$ .  
**Ans.**  $\{y (x^2 - 1)\}/x = \log x + (1/2) \times x^{-2} + c$

# Bernoulli's equation:



An equation of the form  $(dy/dx) + Py = Qy^n$  ... (1A)

where  $P$  and  $Q$  are constants or functions of  $x$  alone (and not of  $y$ ) and  $n$  is constant except 0 and 1, is called a *Bernoulli's differential equation*.

We first multiply by  $y^{-n}$ , thereby expressing it in the form (1) of Art. 2.25

$$y^{-n} (dy/dx) + Py^{1-n} = Q. \quad \dots (2 A)$$

Let  $y^{1-n} = v$  ... (3 A)

Differentiating w.r.t.  $x$ , (3 A) gives  $(1 - n) y^{-n} \frac{dy}{dx} = \frac{dv}{dx}$ , or  $y^{-n} \frac{dy}{dx} = \frac{1}{1 - n} \frac{dv}{dx}$  ... (4 A)

Using (3 A) and (4 A), (2 A) reduces to

$$\frac{1}{1 - n} \frac{dv}{dx} + Pv = Q \quad \text{or} \quad \frac{dv}{dx} + P(1 - n)v = Q(1 - n),$$

which is linear in  $v$  and  $x$ . Its I.F. =  $e^{\int P(1-n)dx} = e^{(1-n)\int Pdx}$  and hence the required solution is

$$v \cdot e^{(1-n)\int Pdx} = \int Q \cdot e^{(1-n)\int Pdx} dx + c, \text{ } c \text{ being an arbitrary constant}$$

$$y^{1-n} e^{(1-n)\int Pdx} = \int Q \cdot e^{(1-n)\int Pdx} dx + c, \text{ using (3A)}$$



# Examples.



**Ex. 1.** Solve  $(dy/dx) + x \sin 2y = x^3 \cos^2 y$ .

**[I.A.S. (Prel.) 2005; I.A.S. 1994; Calcutta 1995; Kanpur 1997; Lucknow 1996]**

**Sol.** Dividing by  $\cos^2 y$ ,  $\sec^2 y (dy/dx) + 2x (\tan y) = x^3$ . ... (1)

Put  $\tan y = v$  so that  $\sec^2 y (dy/dx) = dv/dx$  Hence the above eqn. becomes  $dv/dx + 2xv = x^3$ ,

which is linear in  $v$  and  $x$ . Hence its I.F. =  $e^{\int 2x dx} = e^{x^2}$  and its solution is given by

$$v \cdot e^{x^2} = \int x^3 e^{x^2} dx + c, c \text{ being an arbitrary constant}$$

$$ve^{x^2} = (1/2) \times \int t e^t dt + c, \text{ putting } x^2 = t \text{ and } 2x dx = dt$$

$$= (1/2) \times [t \times e^t - \int (1 \times e^t) dt] + c = (1/2) \times (t e^t - e^t) + c$$

or  $\tan y \cdot e^{x^2} = (1/2) \times e^{x^2} (x^2 - 1) + c$ , as  $v = \tan y$  and  $t = x^2$

or  $\tan y = (1/2) \times (x^2 - 1) + ce^{-x^2}$ , dividing by  $e^{x^2}$

# Examples (Continu...).



**Ex. 2.** Solve  $(dy/dx) = e^{x-y} (e^x - e^y)$ .

[Agra 1995; Delhi Maths (G) 1997;  
Kanpur 1997; Rohilkhand 1997]

**Sol.** Re-writing,  $dy/dx = e^{2x} \cdot e^{-y} - e^x$  or  $dy/dx + e^x = e^{2x} \cdot e^{-y}$ .

Now dividing by  $e^{-y}$ , we get  $e^y (dy/dx) + e^x \cdot e^y = e^{2x}$ .

Putting  $e^y = v$  so that  $e^y (dy/dx) = dv/dx$  we get  $dv/dx + e^x v = e^{2x}$ .

Its I.F. =  $e^{\int P dx} = e^{\int e^x dx} = e^{e^x}$  and the solution is

$$\begin{aligned} v \cdot e^{e^x} &= \int e^{2x} \cdot e^{e^x} dx + c = \int e^x e^{e^x} \cdot e^x dx + c = \int t e^t dt + c, \text{ putting } e^x = t \text{ so that } e^x dx = dt \\ &= \int t \cdot e^t - \int 1 \cdot e^t dt + c = t \cdot e^t - e^t + c = e^t (t - 1) + c \end{aligned}$$

$$\text{i.e., } e^y e^{e^x} = e^{e^x} (e^x - 1) + c \quad \text{or} \quad e^{e^x} (e^y - e^x + 1) = c, \text{ as } v = e^y \text{ and } t = e^x$$



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## Exercise. 8

1.  $\sin y (dy/dx) = \cos y (1 - x \cos y).$
2.  $(dy/dx) + (1/x) \tan y = (1/x^2) \tan y \sin y.$
3.  $(dy/dx) + 1 = e^{x-y}.$
4.  $(dy/dx) - (\tan y)/(1+x) = (1+x) e^x \sec y.$
5.  $(dy/dx) + (1/x) = e^y/x^2.$  [Kerala 2001]
6.  $(x^2 + y^2 + 2x) dx + 2y dy = 0.$

**Ans.**  $\sec y = x + 1 + ce^x$

**Ans.**  $x \operatorname{cosec} y = c + \log x$

**Ans.**  $e^y = ce^{-x} + (1/2) \times e^x$

**[Kanpur 1998; Lucknow 1996]**

**Ans.**  $\sin y = (1+x)(c + e^x)$

**Ans.**  $2x e^{-y} = 1 + 2cx^2$

**Ans.**  $e^x (x^2 + y^2) = c$



# Equations of the First Order But Not of The First Degree:



## Equations of the first order but not of the first degree

**Defintion.** The general first order differential equation of degree  $n > 1$  is

$$P_0 (dy/dx)^n + P_1 (dy/dx)^{n-1} + P_2 (dy/dx)^{n-2} + \dots + P_{n-1} (dy/dx) + P_n = 0, \quad \dots (1)$$

where  $P_0, P_1, P_2, \dots, P_{n-1}, P_n$  are functions of  $x$  and  $y$ .

It is a convention to denote  $dy/dx$  by  $p$  and so (1) becomes

$$P_0 p^n + P_1 p^{n-1} + P_2 p^{n-2} + \dots + P_{n-1} p + P_n = 0.$$

Such equations can be solved by the following methods:

### **Method I. Equations solvable for $p$ :**

Let 
$$P_0 p^n + P_1 p^{n-1} + P_2 p^{n-2} + \dots + P_{n-1} p + P_n = 0 \quad \dots (1)$$

be the given differential equation of the first order and degree  $n > 1$ .

Since (1) is solvable for  $p$ , it can be put in the form

# Equations of the First Order But Not of The First Degree:



$$[p - f_1(x, y)] [p - f_2(x, y)] \dots [p - f_n(x, y)] = 0. \quad \dots (2)$$

Equating each factor of (2) to zero, we obtain  $n$  equations of the first order and the first degree, namely,

$$p = dy/dx = f_1(x, y), \quad p = dy/dx = f_2(x, y), \quad \dots, \quad p = dy/dx = f_n(x, y). \quad \dots (3)$$

Let the solutions of these  $n$  component equations be respectively

$$F_1(x, y, c_1) = 0, \quad F_2(x, y, c_2) = 0, \quad \dots, \quad F_n(x, y, c_n) = 0,$$

which  $c_1, c_2, \dots, c_n$  are the arbitrary constants of integration.

Since all the  $c$ 's can have any one of an infinite number of values, the above solutions (4) will remain general if we replace  $c_1, c_2, \dots, c_n$  by a single arbitrary constant  $c$ . Then the  $n$  solutions (4) can be re-written as

$$F_1(x, y, c) = 0, \quad F_2(x, y, c) = 0, \quad \dots, \quad F_n(x, y, c) = 0.$$

These can be combined into one equation, namely,

$$F_1(x, y, c) F_2(x, y, c), \dots F_n(x, y, c) = 0.$$



# Equations of the First Order But Not of The First Degree:



**Ex. 1** Solve the following differential

(a)  $(dy/dx)^2 - ax^3 = 0.$

(b)  $p^2 - x^5 = 0.$

(c)  $p^3 = ax^4$

**Sol. (a)** Given  $(dy/dx)^2 = ax^3$  so that  $dy/dx = \pm a^{1/2} x^{3/2}.$

Separating variables,  $dy = \pm a^{1/2} x^{3/2} dx.$

Integrating,  $y + c = \pm (2/5) a^{1/2} x^{5/2}$  or  $5(y + c) = \pm 2a^{1/2} x^{5/2}.$

Squaring both sides,  $25(y + c)^2 = 4ax^5$  which is the required solution.

(b) Proceed as in part (a). **Ans.**  $49(y + c)^2 = 4x^7.$

(c) Given  $(dy/dx)^3 = ax^4$  or  $dy/dx = a^{1/3} x^{4/3}$  or  $dy = a^{1/3} x^{4/3} dx.$

Integrating,  $y + c = (3/7) a^{1/3} x^{7/3},$   $c$  being an arbitrary constant.

Cubing,  $343(y + c)^3 = 27ax^7,$  which is the required solution.



# Equations of the First Order But Not of The First Degree:



**Ex. 2** Solve the following differential:

(a)  $x^2p^2 + xyp - 6y^2 = 0$ .

[Delhi Maths (G) 1993]

(b)  $p^2 + (x + y - 2y/x)p + xy + (y^2/x^2) - y - (y^2/x) = 0$

**Sol. (a)** Given  $x^2p^2 + xyp - 6y^2 = 0$  or  $x^2p^2 + 3xyp - 2xyp - 6y^2 = 0$

or  $xp(xp + 3y) - 2y(xp + 3y) = 0$  or  $(xp + 3y)(xp - 2y) = 0$ .

Its component equations are  $x(dy/dx) + 3y = 0$  and  $x(dy/dx) - 2y = 0$

or  $(1/y)dy + 3(1/x)dx = 0$  and  $(1/y)dy - 2(1/x)dx = 0$ .

Integrating,  $\log y + 3 \log x = \log c$ , i.e.,  $yx^3 = c$  and  $\log y - 2 \log x = \log c$ , i.e.,  $y/x^2 = c$

Hence, the general solution is  $(yx^3 - c)(y/x^2 - c) = 0$ ,  $c$  being an arbitrary constant



# Equations of the First Order But Not of The First Degree:



(b) Re-writing,  $p^2 + [\{x - (y/x)\} + \{y - (y/x)\}]p + \{x - (y/x)\} \{y - (y/x)\} = 0$   
or  $[p + \{x - (y/x)\}] [p + \{y - (y/x)\}] = 0.$

Its component equations are  $(dy/dx) + x - (y/x) = 0 \quad \dots (1)$

and  $(dy/dx) + y - (y/x) = 0. \quad \dots (2)$

Re-writing (1),  $(dy/dx) - (1/x) y = -x$ , which is linear equation.

Its I.F. =  $e^{\int (-1/x) dx} = e^{-\log x} = e^{\log x^{-1}} = x^{-1} = 1/x$  and solution is

$$y \times (1/x) = \int (-x)(1/x) dx + c \quad \text{or} \quad y/x = -x + c. \quad \dots (3)$$

Re-writing (2),  $dy/dx = y \{(1/x) - 1\}$  or  $(1/y) dy = \{(1/x) - 1\} dx$

Integrating,  $\log y = \log x - x + \log c$  or  $y = cx e^{-x}.$

Hence, the combined solution of the given equation is

$$(y + x^2 - cx) (y - cx e^{-x}) = 0, \quad c \text{ being an arbitrary constant.}$$

# Equations of the First Order But Not of The First Degree: Exercise-1



*Solve the following differential equations:*

1.  $p^2 - 5p + 6 = 0$  [Delhi Maths (G) 2005] Ans.  $(y - 2x - c)(cy - 3x - c) = 0$
2.  $p^2 + p = 6$  Ans.  $(y + 3x - c)(y - 2x - c) = 0$
3.  $p^2 - p(e^2 + e^{-2}) + 1 = 0$  Ans.  $(y - x e^2 - c)(y + x e^2 - c) = 0$
4.  $yp^2 + (x - y)p - x = 0$  [Delhi B.A. (Prog.) II 2011] Ans.  $(y - x - c)(y^2 + x^2 - c) = 0$
5.  $x^2 p^2 + 3 xyp + 2 y^2 = 0$  [Madras 2005, Delhi Maths (G) 1993] Ans.  $(xy - c)(x^2 y - c) = 0$
6.  $xyp^2 + (3x^2 - 2y^2)p - 6xy = 0$  Ans.  $(y^2 + 3x^2 - c)(y - cx^2) = 0$
7.  $p^3 + 3xp^2 - y^3p^2 - 3xy^3p = 0$  Ans.  $(y - c)(2y + 3x^2 - c)(y^{-2} + 2x - c) = 0$
8.  $x^2 p^3 + y(1 + x^2 y)p^2 + y^3 p = 0$  Ans.  $(y - c)(x - y^{-1} - c)(ye^{-1/x} - c) = 0$
9.  $xyp^2 + (x^2 + xy + y^2)p + x^2 + xy = 0$  Ans.  $(x^2 + y^2 - c)(2xy + x^2 - c) = 0$
10.  $x^2 p^2 - 2xyp + y^2 = x^2 y^2 + x^4$  Ans.  $x + c = \pm \log [(y/x) + \{1 + (y/x)^2\}^{1/2}]$
11.  $3p^2 y^2 - 2xyp + 4y^2 - x^2 = 0$  Ans.  $2x \pm (x^2 - 3y^2)^{1/2} = c$
12.  $y = x \{p + (1 + p^2)^{1/2}\}$  [Kanpur 1996] Ans.  $x^2 + y^2 = cx$
13.  $xyp^2 - (x^2 - y^2)p - xy = 0$  [Kolkata 1994] Ans.  $(y^2 - x^2 - c)(xy - c) = 0$
14.  $4y^2 p^2 + 2pxy(3x + 1) + 3x^3 = 0$  [Delhi 2008] Ans.  $(x^2 + 2y^2 - c)(x^2 + y^2 - c) = 0$
15.  $p^3(x + 2y) + 3p^2(x + y) + p(y + 2x) = 0$  [Delhi B.Sc. (Prog) II 2010]  
Ans.  $(y - c)(x + y - c)(xy + x^2 + y^2 - c) = 0$

# Equations of the First Order But Not of The First Degree:



## Method II. Equations solvable for x:

If the given equation  $f(x, y, p) = 0$  is solvable for  $x$ , we can express  $x$  explicitly in terms of  $y$  and  $p$ . Thus, an equation solvable for  $x$  can be put in the form.  $x = F(y, p)$  ... (1)

Differentiating (1) w.r.t. 'y' and writing  $1/p$  for  $dx/dy$ , we get

$$1/p = \phi(y, p, dp/dy), \quad \dots (2)$$

which is a equation involving two variables  $y$  and  $p$ . Let its solution be

$$\psi(y, p, c) = 0, \text{ } c \text{ being an arbitrary constant.} \quad \dots (3)$$

Eliminating  $p$  between (1) and (3), we get the solution of (1) in the form  $g(x, y, c) = 0$ .

If the elimination of  $p$  between (1) and (3) is not possible, then we solve (1) and (3) to express  $x$  and  $y$  in terms of  $p$  and  $c$  in the form

$$x = f_1(p, c), \quad y = f_2(p, c). \quad \dots (4)$$

These two equations together form the general solution of (1) in the parametric form, the parameter being  $p$ .

Sometimes even the form (4) of the desired solution is not possible. In that case (1) and (3) may be regarded as giving  $x$  and  $y$  in terms of  $p$ , that is, (1) and (3) together are said to form the solution in parametric form.



# Equations of the First Order But Not of The First Degree:



**Ex. 1.** Solve the following differential equations :

(a)  $y = 2px + y^2 p^3$ .

[Delhi B.Sc. (Prog) II 2008; Purvanchal 2007]

(b)  $p^3 - 4xyp + 8y^2 = 0$ .

[Delhi B.Sc. Maths (Prog) II 2007]

(c)  $(2x - b)p = y - ayp^2, a > 0$

[I.A.S. 2001]

**Sol. (a)** Given

$$y = 2px + y^2 p^3 \text{ where } p = dy/dx. \quad \dots (1)$$

Solving (1) for  $x$ ,

$$x = y(1/2p) - y^2(p^2/2). \quad \dots (2)$$

Differentiating (2) w.r.t. 'y' and writing  $1/p$  for  $dx/dy$ , we get

$$\frac{1}{p} = \frac{1}{2p} - \frac{y}{2p^2} \frac{dp}{dy} - \frac{2y p^2}{2} - \frac{y^2}{2} \cdot 2p \frac{dp}{dy}$$

$$\text{or } \frac{1}{2p} + yp^2 + \frac{dp}{dy} \left( \frac{y}{2p^2} + yp^2 \right) = 0 \quad \text{or} \quad p \left( py + \frac{1}{2p^2} \right) + y \frac{dp}{dy} \left( py + \frac{1}{2p^2} \right) = 0$$

$$\text{or } [py + (1/2 p^2)] [p + y (dp/dy)] = 0.$$

\*Neglecting the first factor which does not involve  $dp/dy$ , (3) reduces to

$$p + y (dp/dy) = 0$$

or

$$(1/p) dp + (1/y) dy = 0.$$

# Equations of the First Order But Not of The First Degree:



Integrating,  $\log p + \log y = \log c$  or  $py = c$  or  $p = c/y$ .

Substituting this value of  $p$  in (1), we get the required solution

$$y = (2x) (c/y) + y^2 (c/y)^3 \quad \text{or} \quad y^2 = 2 cx + c^3.$$

(b) Given  $p^3 - 4xyp + 8 y^2 = 0$ , where  $p = dy/dx$ . ... (1)

Solving (1) for  $x$ ,  $x = (1/4y) p^2 + (1/p) (2y)$ . ... (2)

Differentiating (2) w.r.t. 'y' and writing  $1/p$  for  $dx/dy$ , we get

$$\frac{1}{p} = -\frac{p^2}{4y^2} + \frac{2p}{4y} \frac{dp}{dy} + \frac{2}{p} - \frac{2y}{p^2} \frac{dp}{dy} \quad \text{or} \quad \frac{p^2}{4y} - \frac{1}{p} - \frac{dp}{dy} \left( \frac{p}{2y} - \frac{2y}{p^2} \right) = 0$$

$$\left( \frac{p^2}{4y} - \frac{1}{p} \right) - \frac{2y}{p} \frac{dp}{dy} \left( \frac{p^2}{4y} - \frac{1}{p} \right) = 0 \quad \text{or} \quad \left( \frac{p^2}{4y} - \frac{1}{p} \right) \left( 1 - \frac{2y}{p} \frac{dp}{dy} \right) = 0 \quad \dots (3)$$

Neglecting the first factor which does not involve  $dp/dy$ , (3) reduces to

$$1 - (2y/p) (dp/dy) = 0 \quad \text{or} \quad (2/p) dp = (1/y) dy.$$

Integrating,  $2 \log p = \log y + \log c'$  so that  $p^2 = c' y$ . ... (4)

We now proceed to eliminate  $p$  between (1) and (4). Re-writing (1),  $p (p^2 - 4xy) = -8 y^2$

# Equations of the First Order But Not of The First Degree:



Squaring,  $p^2 (p^2 - 4xy)^2 = 64 y^4$  or  $c'y (c'y - 4xy)^2 = 64 y^4$ , using (4) ... (5)

Let  $c' = 4c$ , where  $c$  is new arbitrary constant. Then (5) gives

$$4cy (4cy - 4xy)^2 = 64y^4 \quad \text{or} \quad c (c - x)^2 = y. \quad \dots (6)$$

(c) Solving the given equation for  $x$ , we have  $2x = b + (y/p) - apy. \quad \dots (1)$

Differentiating (1) w.r.t. 'y', we have

$$\frac{2}{p} = \frac{1}{p} - \frac{y}{p^2} \frac{dp}{dy} - ap - ay \frac{dp}{dy} \quad \text{or} \quad \left( \frac{1}{p} + ap \right) \left( 1 + \frac{y}{p} \frac{dp}{dy} \right) = 0$$

Neglecting the first factor which does not involve  $dp/dy$ , we get

$$1 + (y/p) (dp/dy) = 0 \quad \text{or} \quad (1/p) dp + (1/y) dy = 0 \quad \text{so that} \quad py = c.$$

Putting  $p = c/y$  in the given equation, the required solution is

$$(a c^2/y) + (2x - b) (c/y) - y = 0 \quad \text{or} \quad ac^2 + (2x - b) c - y^2 = 0.$$

# Equations of the First Order But Not of The First Degree:



**Ex. 2.** Solve  $y = 2px + p^2y$ . [Kakatiya 2003, Purvanchal 2007 ; Delhi (Prog.) II 2011]

**Sol.** Solving for  $x$ , we get  $2x = -py + y/p$ . ... (1)

Differentiating (1) w.r.t.  $y$ , we get  $\frac{2}{p} = -p - y \frac{dp}{dy} + \frac{1}{p} - \frac{y}{p^2} \frac{dp}{dy}$ , as  $\frac{dx}{dy} = \frac{1}{p}$

or  $p + 1/p = -y (dp/dy) (1 + 1/p^2)$  or  $(1 + 1/p^2) + y (dp/dy) (1 + 1/p^2) = 0$

or  $(1 + 1/p^2) [p + y (dp/dy)] = 0$ .

Omitting the first factor which leads to a singular solution, we get

$p + y (dp/dy) = 0$  or  $(1/p) dp + (1/y) dy = 0$

Integrating  $\log p + \log y = \log c$  or  $py = c$ . ... (2)

To eliminate  $p$  between (1) and (2), first solve (2) for  $p$ . So  $p = c/y$ . Putting this value of  $p$  in (1), we get  $2x = -c + y^2/c$  or  $2xc - y^2 + c^2 = 0$ .



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# Equations of the First Order But Not of The First Degree: Exercise-2



*Solve the following differential equations*

1.  $p^3y^2 - 2px + y = 0$  [Delhi B.Sc. (Prog) II 2011]

Ans.  $y^2 = 2cx - c^3$

2.  $yp^2 - 2xp + y = 0$

Ans.  $y^2 = 2cx - c^2$

3. (i)  $y - 2xp + ayp^2 = 0$  (ii)  $y - 2xp + yp^2$  (I.A.S. 2008)

Ans. (i)  $y^2 = 2cx - ac^2$  (ii)  $y = 2cx - c^2$

4.  $4(xp^2 + yp) = y^4$

Ans.  $y = 4c(xyc + 1)$

5.  $y = 3px + 6p^2y^2$  [Delhi Maths (H) 2001; Delhi Maths (G) 2000]

Ans.  $y^3 = 3cx + 6c^2$

6.  $p^2 - 2xp + 1 = 0$

Ans.  $x = (p/2) + (1/2p), y = (p^2/4) - (1/2) \times \log p + c$

7.  $xp^3 = a + bp$

Ans.  $x = a/p^3 + b/p^2, y = (3a/2p^2) + (2b/p) + c$

8.  $xp^3 - p^2 - 1 = 0$

Ans.  $x = p^{-1} + p^{-3}, y = (3/2) \times p^{-2} - \log p + c$

9.  $p = \tan \{x - p/(1 + p^2)\}$  or  $p \cot \{x - p/(1 + p^2)\} = 1$

Ans.  $x = \{p/(1 + p^2)\} + \tan^{-1} p, y = c - (p^2 + 1)^{-1}$



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# Equations of the First Order But Not of The First Degree:



## Method III. Equations solvable for y:

If the given equation  $f(x, y, p) = 0$  is solvable for  $y$ , we can express  $y$  explicitly in terms of  $x$  and  $p$ . Thus, an equation solvable for  $y$  can be put in the form  $y = F(x, p)$ . ... (1)

Differentiating (1) w.r.t.  $x$  and writing  $p$  for  $dy/dx$ , we get  $p = \phi(x, p, dp/dx)$ , ... (2)  
which is an equation involving two variables  $x$  and  $p$ . Let its solution be

$$\psi(x, p, c) = 0, c \text{ being an arbitrary constant.} \quad \dots (3)$$

Eliminating  $p$  between (1) and (3), the solution of (1) is in the form  $g(x, y, c) = 0$ .

If the elimination of  $p$  between (1) and (3) is not possible, then we solve (1) and (3) to express  $x$  and  $y$  in terms of  $p$  and  $c$  in the form

$$x = f_1(p, c), \quad y = f_2(p, c). \quad \dots (4)$$

These two equations together form the general solution of (1) in the parametric form, the parameter being  $p$ .

Sometimes even the form (4) of the desired solution is not possible. In that case (1) and (3) may be regarded as giving  $x$  and  $y$  in terms of  $p$ , that is, (1) and (3) together are said to form the solution in parametric form.

# Equations of the First Order But Not of The First Degree:



**Remark 1.** In some problems (2) can be expressed as  $\phi_1(x, p) \phi_2(x, p, dp/dx) = 0$ . ... (5)

In such cases we ignore the first factor  $\phi_1(x, p)$  which does not involve  $dp/dx$  and proceed with  $\phi_2(x, p, dp/dx) = 0$  as discussed in Art. 4.6.

**Remark 2.** If instead of ignoring the factor  $\phi_1(x, p)$ , we eliminate  $p$  between (1) and  $\phi_1(x, p) = 0$ , we obtain an equation involving no constant  $c$ . This is known as *singular solution* of (1) and we shall discuss it later on in this chapter. Singular solution will be obtained only if asked to do so in a given question. For details refer part II of this chapter.

# Equations of the First Order But Not of The First Degree:



## First special case of Method III. Lagrange's equation

The differential equation of the form  $y = x F(p) + f(p)$  ... (6)

is known as *Lagrange's equation*. To solve it we proceed as in Art. 4.6.

Differentiating (6) w.r.t. 'x', we have  $p = F(p) + x F'(p) (dp/dx) + f'(p) (dp/dx)$

$$\text{or } p - F(p) = \frac{dp}{dx} [x F'(p) + f'(p)] \quad \text{or} \quad \frac{dx}{dp} = \frac{x F'(p) + f'(p)}{p - F(p)}$$

$$\text{or } \frac{dx}{dp} - \frac{F'(p)}{p - F(p)} x = \frac{f'(p)}{p - F(p)},$$

which is linear equation in  $x$  and  $p$  and can be solved by usual method to give a relation of the form

$$x = \phi(p, c) \quad \dots (7)$$

We now eliminate  $p$  between (6) and (7) to get the required solution

It  $p$  cannot be eliminated, then putting the value of  $x$  in (6), we get

$$y = \phi(p, c) F(p) + f(p). \quad \dots (8)$$

Then (7) and (8) together form the required solution in parametric form,  $p$  being regarded as parameter.

# Equations of the First Order But Not of The First Degree:



## **Second special case of method III. Equation that do not contain $x$ .**

If given equation does not contain  $x$ , then it can be put in the form  $f(y, p) = 0$ . ... (9)

If (9) is solvable for  $p$ , it can be put in the form  $p = F(y)$  or  $dy/dx = F(y)$ , which can be easily integrated to give the required solution.

If (9) is solvable for  $y$ , it can be put in the form  $y = G(p)$ , which is of the form (1) of Art. 4.6 and so it can be solved as before.



# Equations of the First Order But Not of The First Degree:



**Ex. 1.** Solve (a)  $y = 3x + \log p$ .

(b)  $y = x \{p + (1 + p^2)^{1/2}\}$ .

[Kanpur 2009]

**Sol.** (a) Given  $y = 3x + \log p$ , where  $p = dy/dx$ . ... (1)

Differentiating (1) w.r.t. 'x' and writing  $p$  for  $dy/dx$ , we get

$$p = 3 + (1/p) (dp/dx) \quad \text{or} \quad p(p - 3) = (dp/dx)$$

or 
$$dx = \frac{dp}{p(p - 3)} = \frac{1}{3} \left[ \frac{1}{p - 3} - \frac{1}{p} \right] dp, \text{ resolving into partial fractions.}$$

Integrating,  $x = (1/3) [\log(p - 3) - \log p] - (1/3) \log c$ ,  $c$  being an arbitrary constant

or 
$$\log \left[ \frac{p - 3}{pc} \right] = 3x \quad \text{or} \quad \frac{p - 3}{pc} = e^{3x} \quad \text{or} \quad p = \frac{3}{1 - ce^{3x}}$$

Putting this value of  $p$  in (1), the required solution is

$$y = 3x + \log \{3/(1 - ce^{3x})\}, c \text{ being an arbitrary constant}$$



# Equations of the First Order But Not of The First Degree:



(b) Given  $y = x \{p + (1 + p^2)^{1/2}\}$ , where  $p = dy/dx$ . ... (1)

Differentiating (1) w.r.t. 'x' and writing  $p$  for  $dy/dx$ , we get

$$p = p + (1 + p^2)^{1/2} + x \left[ 1 + \frac{1}{2}(1 + p^2)^{-1/2} \times 2p \right] \frac{dp}{dx}$$

$$\text{or } \frac{x[p + (1 + p^2)^{1/2}]}{(1 + p^2)^{1/2}} \frac{dp}{dx} = -(1 + p^2)^{1/2} \quad \text{or} \quad \left[ \frac{1}{2} \cdot \frac{2p}{1 + p^2} + \frac{1}{(1 + p^2)^{1/2}} \right] dp = -\frac{dx}{x}.$$

$$\text{Integrating,} \quad (1/2) \times \log(1 + p^2) + \log[p + (1 + p^2)^{1/2}] = \log c - \log x$$

$$\log(c/x) = \log(1 + p^2)^{1/2} + \log[p + (1 + p^2)^{1/2}]$$

$$\text{or } c/x = (1 + p^2)^{1/2} [p + (1 + p^2)^{1/2}] \quad \text{or} \quad c/x = (1 + p^2)^{1/2} (y/x), \text{ by (1)}$$

$$\text{or } c = y(1 + p^2)^{1/2} \quad \text{or} \quad c^2 = y^2(1 + p^2) \quad \text{or} \quad c^2 - y^2 = y^2 p^2. \quad \dots (2)$$

$$\text{Multiplying both sides of (1) by } y, \quad y^2 = xyp + xy(1 + p^2)^{1/2}$$

$$\text{or } y^2 = xyp + cx, \text{ since from (2)} \quad y(1 + p^2)^{1/2} = c$$

$$\text{or } y^2 - cx = xyp \quad \text{or} \quad (y^2 - cx)^2 = x^2 y^2 p^2$$

$$\text{or } y^4 - 2cxy^2 + c^2 x^2 = x^2 y^2 p^2 \quad \text{or} \quad y^4 - 2cxy^2 + c^2 x^2 = x^2 (c^2 - y^2), \text{ by (2)}$$

$$\text{or } y^4 - 2cxy^2 + x^2 y^2 = 0 \quad \text{or} \quad x^2 + y^2 - 2xc = 0$$

# Equations of the First Order But Not of The First Degree:



**Ex. 2** Solve :  $yp^2 - 2xp + y = 0$

**Sol.** Solving for  $y$ ,  $y = (2px) / (1 + p^2)$ . ... (1)

Differentiating (1) w.r.t. 'x' and writing  $p$  for  $dy/dx$ , we get

$$p = \frac{(1 + p^2)[2p + 2x(dp/dx)] - 2px \cdot (2p)(dp/dx)}{(1 + p^2)^2}$$

or  $p(1 + p^2)^2 = 2p(1 + p^2) + 2x(dp/dx)\{1 + p^2 - 2p^2\}$

or  $p(1 + p^2)(1 + p^2 - 2) = -2x(p^2 - 1)(dp/dx)$  ... (2)

or  $(p^2 - 1)[p(1 + p^2) + 2x(dp/dx)] = 0$ .

Neglecting the first factor which does not involve  $dp/dx$ , (2) reduces to

$$p(1 + p^2) + 2x \frac{dp}{dx} = 0 \quad \text{or} \quad \frac{dx}{x} + \left[ \frac{2}{p} - \frac{2p}{1 + p^2} \right] dp = 0,$$

$$\therefore \log x + 2 \log p - \log(1 + p^2) = \log c \quad \text{or} \quad x = c(1 + p^2)/p^2 \quad \dots (3)$$

Putting this value of  $x$  in (1), we get  $y = 2c/p$ . ... (4)

(2) and (4) together form the solution in parametric form,  $p$  being treated as parameter.

# Equations of the First Order But Not of The First Degree: Exercise-3



*Solve following differential equations :*

1.  $y = 2px + p^4x^2$  [Delhi Maths (G) 1998] **Ans.**  $(y - c^2)^2 = 4cx$
2.  $xp^2 - 2yp + ax = 0$  [Delhi Maths (H) 2006; G.N.D.U. Amritsar 2010] **Ans.**  $c^2x^2 - 2yc + a = 0$
3.  $y = 2px - xp^2$  [Delhi B.A. (Prog) II 2007, 10] **Ans.**  $(y + c)^2 = 4cx$
4.  $xp^2 - 3yp + 9x^2 = 0$  **Ans.**  $3cy = c^2x^3 + 9$
5.  $p^2 + p = y/x$  or  $x(p^2 + p) = y$ . **Ans.**  $x = (c/p^2) e^{1/p}$ ,  $y = c(1 + 1/p) e^{1/p}$
6.  $y = p^2 + x$ . **Ans.**  $x = 2p + 2 \log(p - 1) + c$ ,  $y = p^2 + 2p + 2 \log(p - 1) + c$
7.  $y = p^3 + x$ . **Ans.**  $x = 3p^2/2 + 3p - 3 \log(p - 1) + c$ ,  
 $y = p^3 + 3p^2/2 + 3p + 3 \log(p - 1) + c$
8.  $y - px + x - (y/b) = a$ . **Ans.**  $x = c(p - b)^{-b}$ ,  $y = \{b/(b - 1)\} \times \{c(p - 1)(p - b)^{-b} + a\}$
9.  $y = p + xp^2$  [Agra 1995] **Ans.**  $x = (\log p - p + c) / (p - 1)^2$ ,  $y = p \{1 - p(2 - \log p)\} / (p - 1)^2$
10.  $x^2 + p^2x = yp$ . **Ans.**  $x = cp^{1/2} - (p^2/3)$ ,  $y = \{c - (1/3)p^{3/2}\} \times \{c + (2/3)p^{3/2}\}$
11.  $x = yp - p^2$  **Ans.**  $x = \{p/(1 + p^2)^{1/2}\} (c - \sin^{-1} p)$ ,  $y = \{1/(1 + p^2)^{1/2}\} (c - \sin^{-1} p) + p$
12.  $x + py = p^3$  [Delhi B.Sc. (Prog) II 2009; Delhi Maths (G) 2000, Delhi Maths (H) 1999] **Ans.**  $x = 2p + cp(p^2 + 1)^{-1/2}$ ,  $y = p^2 - c(p^2 + 1)^{-1/2} - 2$

# Equations of the First Order But Not of The First Degree:



## Method IV : Equations in Clairaut's form:

### Clairaut's equation. Definition.

An equation of the form  $y = px + f(p)$  is known as *Clairaut's equation*.

**General solution of Clairaut's equation.** *To show that the general solution of Clairaut's equation  $y = px + f(p)$  is  $y = cx + f(c)$  which is obtained by replacing  $p$  by  $c$ , where  $c$  is an arbitrary constant*



# Equations of the First Order But Not of The First Degree:



**Ex. 1.** Solve the following differential equations :

(a)  $y = px + (1 + p^2)^{1/2}$

[Kanpur 1995]

(b)  $y = px + ap(1 - p)$

[Agra 2006]

(c)  $y = px + (a/p)$  or  $y = x(dy/dx) + a(dx/dp)$

[Agra 2005]

(d)  $y = x(dy/dx) + e^{dy/dx}$  or  $y = xp + e^p$

(e)  $y = x(dy/dx) + (dy/dx)^2$  or  $y = px + p^2$

[I.A.S. (Prel.) 1997]

**Sol.** (a) The given equation is in Clairaut's form  $y = px + f(p)$ . So replacing  $p$  by  $c$  its general solution is  $y = cx + (1 + c^2)^{1/2}$ ,  $c$  being an arbitrary constant.

(b) Proceed as in part (a).

**Ans.**  $y = cx + ac(1 - c)$

(c) Proceed as in part (a).

**Ans.**  $y = cx + (a/c)$

(d) Writing  $p$  for  $dy/dx$ , we get  $y = xp + e^p$  which is in Clairaut's form  $y = px + f(p)$ . So replacing  $p$  by  $c$  its general solution is  $y = cx + e^c$ .

(e) Writing  $p$  for  $dy/dx$ , the given equation reduces to  $y = px + p^2$ , which is in Clairaut's form  $y = px + f(p)$ . Replacing  $p$  by  $c$ , the required solution is  $y = cx + c^2$ ,  $c$  being an arbitrary constant



# Equations of the First Order But Not of The First Degree:



**Ex. 2.** Solve the following differential equations :

(a)  $p = \log (px - y).$

**I.A.S. (Prel.) 2005]**

(b)  $p = \tan (px - y).$

**[Jabalpur 1993, Poona 1991]**

(c)  $\sin px \cos y = \cos px \sin y + p$

**[I.A.S. (Prel.) 2005]**

(d)  $(y - px)^2 / (1 + p^2) = a^2$

(e)  $p^2 (x^2 - a^2) - 2 pxy + y^2 - b^2 = 0$

**[G.N.D.U. Amritsar 2010]**

(f)  $y^2 + x^2 (dy/dx)^2 - 2xy (dy/dx) = 4 (dx/dy)^2$

**Sol.** (a) Given  $p = \log (px - y)$  so that  $e^p = px - y$  or  $y = px - e^p$ , which is in Clairaut's form. So replacing  $p$  by  $c$ , the required general solution is

$$y = cx - e^c, c \text{ being an arbitrary constant.}$$

(b) Given  $p = \tan (px - y)$  so that  $\tan^{-1} p = px - y$  or  $y = px - \tan^{-1} p$ , which is in Clairaut's form. So replacing  $p$  by  $c$ , the required solution is

$$y = cx - \tan^{-1} c, c \text{ being an arbitrary constant}$$

# Equations of the First Order But Not of The First Degree:



(c) Given  $\sin px \cos y = \cos px \sin y + p$  or  $\sin px \cos y - \cos px \sin y = p$   
or  $\sin (px - y) = p$  or  $px - y = \sin^{-1} p$  or  $y = px - \sin^{-1} p$ , which is in Clairaut's form.  
So replacing  $p$  by  $c$ , the required general solution is

$$y = cx - \sin^{-1} c, c \text{ being an arbitrary constant.}$$

(d) Given  $(y - px)^2 / (1 + p^2) = a^2$  or  $y - px = \pm a (1 + p^2)^{1/2}$   
or  $y = cx \pm a (1 + p^2)^{1/2}$ , which is in Clairaut's form.

Replacing  $p$  by  $c$ , its general solution is  $y = cx \pm a (1 + c^2)^{1/2}$ .

or  $y - cx = \pm a (1 + c^2)^{1/2}$  or  $(y - cx)^2 = a^2 (1 + c^2)$ , on squaring.



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# Equations of the First Order But Not of The First Degree: Exercise-4



*Solve the following differential equations, taking  $p = dy/dx$*

1.  $y = px + p^n$

**Ans.**  $y = cx + c^n$

2.  $y = px + p - p^2$  [Kanpur 2011]

**Ans.**  $y = cx + c - c^2$

3.  $y = px + \log p$

**Ans.**  $y = cx + \log c$

4.  $y = px + \sin^{-1} p$

**Ans.**  $y = cx + \sin^{-1} c$

5.  $y = x (dy/dx) + (dy/dx)^3$

**Ans.**  $y = cx + c^3$

6.  $y = px + a \tan^{-1} p$

**Ans.**  $y = cx + a \tan^{-1} c$

7.  $px - y + p^3 = m^3/p^3$

**Ans.**  $y = cx + c^3 - (m^3/c^3)$

8.  $y = px + (a^2p^2 + b^2)^{1/2}$

**Ans.**  $y = cx + (a^2c^2 + b^2)^{1/2}$

9.  $xp^2 - yp + a = 0$  or  $y = px + a/p$  [Purvanchal 2007]

**Ans.**  $y = cx + (a/c)$

10.  $\sin(y - px) = p$  [Delhi Maths (Hons.) 2005]

**Ans.**  $y = cx + \sin^{-1} c$



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